

LEA LABS · APPLIED AI

Applied AI Foundations & Literacy

Learn what AI can do, how it learns from data, and how to use it with curiosity, care, and evidence.

YOU ARE THE CRITICAL THINKER



From Rules to Creation

Traditional Code

Humans write explicit rules.
The computer applies these rules to data to produce answers.

Rules + Data = Answers

Machine Learning

Humans provide data and the correct answers. The computer figures out the rules.

Data + Answers = Rules

Generative AI

The computer learns deep patterns from massive data to create entirely new content.

Patterns = New Content

Data Becomes Signal

Examples & Labels

AI models learn by observing thousands of examples. Labels act as the answer key, telling the model exactly what it is looking at and turning raw data into a learning opportunity.

Features & Patterns

Instead of memorizing exact images or texts, the model identifies underlying features (like shapes, colors, or word structures) and finds patterns to make predictions on new, unseen data.



How a Model Learns

Finding patterns instead of memorizing answers.

1. TRAINING

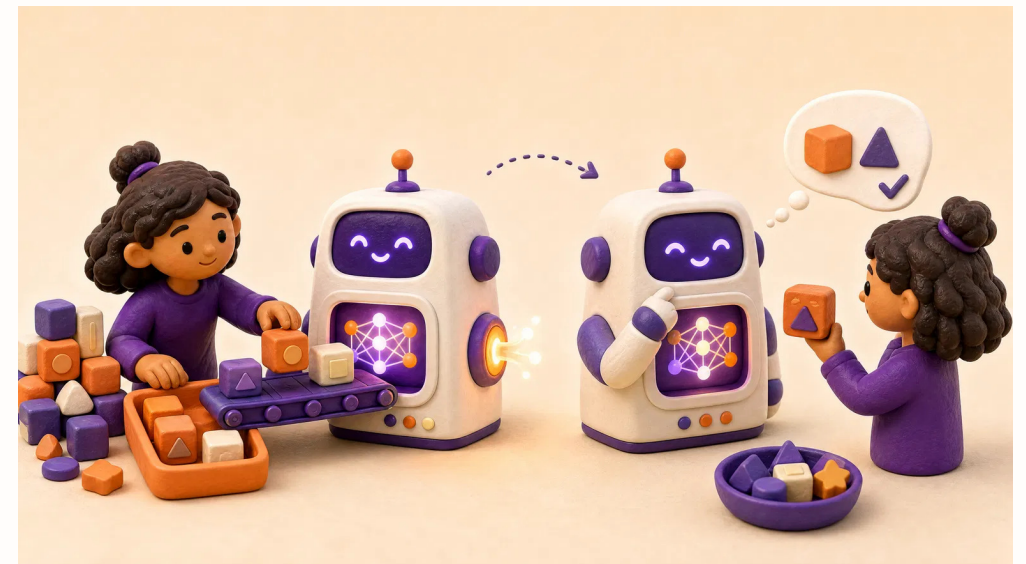
The model looks at thousands of examples to find underlying rules and patterns. It guesses, checks its answer, and adjusts its internal connections.

2. TESTING

We hide some data during training. Later, we test the model on this unseen data to prove it actually learned the concept, not just the specific examples.

THE OVERFITTING TRAP

If a model memorizes the training data perfectly, it will fail in the real world. True learning means generalizing to new, unexpected situations.



Classification in **Everyday Life**

Probabilities & Thresholds

Models don't just say "Spam" or "Not Spam". They predict a probability (e.g., 85% likely to be spam). We set a **threshold** to turn that probability into a final decision.

Precision vs. Recall

Precision: When the model says it's spam, how often is it right? (Avoids false alarms).

Recall: Out of all actual spam, how much did the model catch? (Avoids missing things).

The Confusion Matrix (Spam Filter)

	Actual Spam	Actual Inbox
Predicted Spam	True Positive Spam correctly caught	False Positive Good email sent to spam
Predicted Inbox	False Negative Spam sneaks into inbox	True Negative Good email stays in inbox

Generative AI

TOKENS, CONTEXT, AND WHY FLUENT OUTPUT IS NOT PROOF

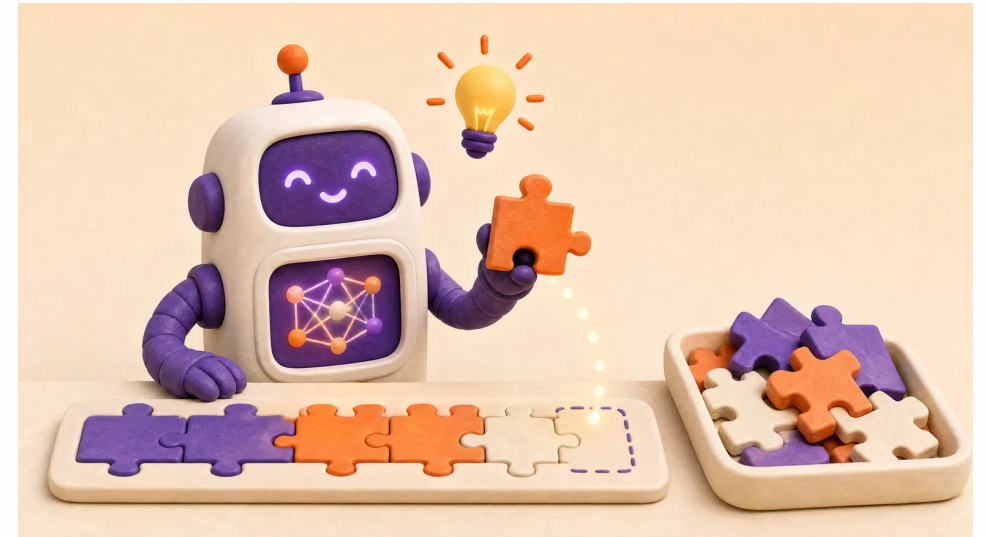
■ Predicting the Next Token

Models predict the most likely next token from patterns in training data.

■ Context Windows

Responses depend on your prompt and a limited memory window, not independent understanding.

Fluency ≠ Fact: Confident wording can still be false — verify key claims.



Prompting as Specification

1. Goal

Specify the exact output and format.

2. Context

Provide only the essential background.

3. Constraints

List rules (tone, length, things to avoid).

4. Examples (Few-Shot)

Give 1–2 concise examples.

5. Checks

Ask for verification or brief reasoning.

BEFORE: VAGUE & UNPREDICTABLE

"Write a summary of this article."

AFTER: CLEAR SPECIFICATION

"Act as a technical editor: summarize the text into exactly three bullet points; use bold for each main concept; avoid jargon. Before finalizing, list the three main concepts you identified."

The Failure Lab

Hallucination

The model generates highly confident but entirely fabricated information, presenting falsehoods as undeniable facts.

Bias & Prejudice

The model replicates or amplifies historical prejudices, stereotypes, and inequalities present in its training data.

Data Leakage

The model inadvertently exposes sensitive, proprietary, or private information that was included in its training or context window.

Automation Bias

Humans over-trust the AI's output, failing to verify results and blindly accepting algorithmic decisions over human judgment.

Human-Centered Evaluation

1. Test Cases

Define a structured set of expected inputs, including standard requests, edge cases, and boundary conditions, before deploying the prompt.

2. Rubrics

Establish clear scoring criteria based on accuracy, tone, format adherence, and safety to objectively measure the model's performance.

3. Red-Team Prompts

Actively attempt to break the system by feeding it adversarial inputs, confusing context, or malicious requests to expose vulnerabilities.

THE STANDARD:

Never trust a prompt based on a single successful output. Always demand evidence-based review across multiple trials.

Responsible AI

Human Accountability **GOVERN**

Algorithms do not take responsibility; humans do. Establish clear oversight for AI-assisted decisions.

Transparency & Privacy **MAP**

Understand what data the model uses, how it processes it, and ensure sensitive information is never exposed.

Fairness & Bias **MEASURE**

Actively test for demographic, cultural, or systemic biases to ensure equitable outcomes for all users.

Safety & Security **MANAGE**

Implement guardrails, fallback mechanisms, and continuous monitoring to mitigate risks in deployment.



Knowledge Check

Scenario 01: A lawyer uses an AI assistant to draft a brief. The AI confidently cites three past legal cases that support the argument, complete with dates and docket numbers. Later, the judge discovers none of those cases actually exist.

HALLUCINATION

Scenario 02: A company implements an AI resume screener. After a month, HR notices the system systematically downgrades applicants who attended specific regional universities, reflecting historical hiring patterns in the training data.

BIAS & FAIRNESS

Scenario 03: A junior developer copies a block of unreleased, proprietary company code and pastes it into a public consumer chatbot, asking it to "find the bug in this logic."

DATA LEAKAGE

Lab Launch: Build an AI Helper

DELIVERABLES

- A complete prompt specification (Goal, Context, Constraints, Examples, Checks).
- Documentation of data sources and potential privacy risks.
- A human-centered review protocol for the outputs.

SUCCESS CRITERIA

- The prompt consistently formats output exactly as specified.
- At least 3 distinct edge cases have been defined and tested.
- One potential failure mode (e.g., bias or hallucination) is identified and mitigated.

EXTENSION

RED-TEAM CHALLENGE

Write a malicious or highly confusing prompt designed specifically to break your own AI helper's constraints. Document how it failed.